

PORTMAN GPS/GPRS SECURITY SYSTEM PROTOCOL

FOR GT3101MG-Tunav (V1.1)

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Portman GPRS tracking protocol

Introduction

This document describes the Portman car alarm / tracking system protocol. Our initial target is to utilize GPS tracking features and also combine Portman's alarm functionalities.

Any of the trigger report can be customized to being sent or not via PC program or from over-the-air command, which makes this device very flexible.

There are 3 main groups: (1) Keep-alive procedure, (2) Automatic report format and (3) server control command.

In order to make the UNIT work under both GPRS and SMS, the server command may need to send the control commands via GPRS or SMS, except some omissions explained in the following sections.

Summary of the parameter settings:

- (1) Time/interval: in sec. (no decimal)
- (2) Speed: km/h (no decimal)
- (3) Distance: in km. (with or without decimal)
- (4) Course change: degree (no decimal)
- (5) Location coordinate: 21 fixed digits, e.g. N2446.5321E12120.4231

Pocket header:

- (1) Sent by the UNIT: \$EXT,[ID] (the UNIT will always send its ID right after the header '\$EXT')
- (2) Sent by the server: &&[ID]

Pocket end byte:

[CR][LF] (0x0D 0x0A)

If it is a acknowledge packet from the UNIT, the end bye will have this format:

'OK [CR][LF]'

1. Keep Alive Packet.

The UNIT will automatically send this packet to the server every 'x' seconds in order to maintain connection in UDP or TCP mode. It can be programmed outside from the PC setup program, SMS command, or defined by the server. By default, the time interval is 300 seconds (5 minutes). When the server receives the packet, it will return it with the 'ID'.

The Keep-alive Packet will use the following format:

Sent by the UNIT

%%[ID],[Counter] [CR][LF]

Return by the SERVER

&&[ID],[Counter] [CR][LF]

For example	Sent by the UNIT	Return by the SERVER
ASCII	1st send: %%PTM01,0[CR][LF]	&&PTM01,0[CR][LF]
HEX	25 25 50 54 4D 30 31 2C 30 0D 0A	26 26 50 54 4D 30 31 2C 30 0D 0A
ASCII	2nd send: %%PTM01,1[CR][LF]	&&PTM01,1[CR][LF]
HEX	25 25 50 54 4D 30 31 2C 31 0D 0A	26 26 50 54 4D 30 31 2C 31 0D 0A

%% and && are the packet headers

[CR] [LF] (0X0D, 0X0A) is the end byte.

[ID]: The UNIT ID in alphanumeric string, which can be as long as 15 characters.

For example, PTM01, or 1234567890...

[Counter]: 1 byte in ASCII from 0 to 9. UNIT will automatically increase '1' at every time.

For example:

1st send: %%PTM01,0[CR][LF]

Back: &&PTM01,0[CR][LF]

2nd send: %% PTM01,1[CR][LF]

Back: &&PTM01,1[CR][LF]

...

.. send: %% PTM01,9[CR][LF]

Back: &&PTM01,9[CR][LF]

.. send: %% PTM01,0[CR][LF]

Back: &&PTM01,0[CR][LF]

2. UNIT Report Packet.

The UNIT will regularly send the message to the server and the standard report packet will use the following format:

\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[ID Card number],[Status],[Event],CFG:[Message][CR][LF]

DESCRIPTION:

‘**\$EXT**’ is the header and ‘[CR][LF]’ is the end byte.

[ID]: The UNIT ID in alphanumeric string, which can be as long as 15 characters.

For example, PORTMAN00001, or 1234567890...

[GPS Valid]: This string will be only L or A or B or C

‘A’ means that the GPS is fixed in 3D mode;

‘B’ means that the GPS is fixed in 2D mode;

‘L’ means that the GPS is not fixed, and the location in GPS is “Last Known”.

‘C’ means that the unit is fixed in LBS mode (Location-Based Services, used GSM module), when the unit can't receive GPS

signal

If it is L, then the current GPS data is not valid. The report will be represented with the last known GPS data.

[Date & Time]: The string contains the date and time in GPS data. (GMT Greenwich Mean Time) Fixed length only includes 12 digits.

The format is:

Year month day hour minute second

For example, the string 040315131415 means:

The date is “Year 2004, Month March, Day 15 “
and the time is “13:14:15 ”

[Loc]: The string contains the GPS location.

For example, the string N2446.5321E12120.4231 means

“North 24 degrees 46.5321 minutes

“East 121 degrees 20.4231 minutes”

Or S2446.5281W01234.5678 means

“South 24 degrees 46. 5281 minutes” = “South 24.7755 degrees” = “South 24 degrees 46 minutes 31.69 seconds”

“West 12 degrees 34.5678 minutes” = “West 12.57613 degrees” = “West 12 degrees 34 minutes 34.07 seconds”

[Speed]: The string provides the GPS speed as “ kilometers per hour” format. NON-fixed length and the value is from 0 to 999.

For example, 20 means 20 km per hour. 1 means 1 km/h, and 110 means 110km/h.

[Dir]: The string provides the GPS direction in degrees. And this value is between 0 and 359 degree. (no decimal)

For example, 311 means 311 degree to the north. 0 means 0 degree to the north.

[Temp]: The string provides the data

Format	Explanation	For example	
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NA	It means Ineffective	NA	There is nothing on AD port.
Cxx	It means current temperature is how much "xx" value is -40 ~ 200 (Celsius) If "xx" is "NA" then it means there is no temperature sensor	C-21	The current temperature is minus 21 degrees Celsius.

[Fuel]: reservation (NA : It means ineffective)

Fxx	It means current fuel capacity is how much "xx" value is 00 ~ 100 (percent) If "xx" is "NA" then it means there is no fuel sensor	F85	The current fuel capacity is 85%.
Hxx:xx	Hxx:It means fuel sensor id, "xx" fuel hight (millimetre) If "NA" then it means there is no fuel sensor	H261:125.6	H261: means fuel sensor id is 261, 125.6: means fuel hight is 125.6 millimetre
Hxx:xx:xx:xx	Hxx:It means fuel sensor 1 id, "xx" fuel sensor 1 hight (millimetre) xx:It means fuel sensor 2 id, "xx" fuel sensor 2 hight (millimetre)	H395:319.3:272:234.3	H395: means fuel sensor 1 ID is 395 319.3: means fuel senosr 1 hight is 319.3 (millimetre) 272: means fuel sensor 2 ID is 272 234.3: means fuel sensor 2 hight is 234.3(millimetre)

[number of satellite]: Satellite number: 0~12

[GSM signal]: GSM signal quality: 0~31.

[mileage]: Distance: in kilometre.

[ID Card number]:Current card ID number

For example:

The string 1492319192 means "ID CARD Number is 1492319192"

The string NA means " means the unit does not include a ID CARD

[Status]: We use eight ASCII characters to represent 32-bit binary number. Each bit represents a flag in the unit's status register. The status string will be represented in HEX for each set of the byte. To display a four-byte string, there will be 8 digits string

Status bit	Meanings
0	Panic yes/no-----no=0
1	
2	UNIT is arming(on)/disarming(off)
3	
4	Anti-car jack function on/off
5	car moving on/off (stop) ----- off(stop)=0
6	Start Kill on/off
7	
8	Store forward report? yes/no ----- no=0
9	
10	
11	
12	Fuel the defined lowest fuel level? Yes/no-----no=0
13	
14	
15	
16	Door opened? yes/no ----- no=0
17	ACC on /off ----- off=0
18	
19	Sensor triggered? yes/no ----- no=0
20	
21	Low battery? yes/no ----- no=0
22	Over speed? yes/no ----- no=0
23	Power on/off-----off=0
24	Engine running? yes/no ----- no=0
25	
26	
27	

28	
29	
30	
31	

Example:

When the server receives the status of string (in HEX) '01020120', it can be translated as:

01 means : >>> 00000001

Engine is on

02 means : >>>00000010

INPUT2 (ACC) are on

01means : >>>00000001

store forward report

20 means : >>> 00100000

Car is moving

[Event]: Only one event will be sent at a time.

[Message]: Report configuration information and text

Format: [any text message] |

CFG: xxxx,xxxx |

No starting character for the plain text, but the last character '|' will need being added.

The configuration message will begin with 'CFG:' and end with the last character '|'.

At the moment, we will use this field to transmit text info from the external terminal and also the command configuration parameters.

For GT3100:

When GT3100 try to send a un-delivered report, CFG message will as below:

\$EXT,GT3100,A,130401025807,N2247.5221E11402.7513,012,244,NA,NA,8,15,6571.46,NA,47022000,406,CFG:Y06,GT3100_B_PTM_5.2.7_1.3|

CFG:Y06,GT3100_B_PTM_5.2.7_1.3 is version

Here is the list for GT3101MG CFG information

Event number	Event function	CFG Parameter list	
41	Add fuel report	CFG:[fuel sensor id]:[oil height height]/[oil height]	[fuel sensor id]:fuel sensor number [oil height]:Before refueling oil height [oil height]:After refueling oil height
42	Steal fuel report	CFG:[fuel sensor id]:[oil height height]/[oil height]	[fuel sensor id]:fuel sensor number [oil height]:Stealing the oil before oil height [oil height]:after Steal oil height
268	ID Card information report	CFG:[Id card number]	[Id card number]:
269	Invalid ID Card information report	CFG:[Id card number]	[Id card number]:
276	Update ID card report	CFG:[Id card number]	[Id card number]:
873	Engine idle time report	CFG:[idle time]	for obd/canbus
880	diagnostic trouble code report	CFG:DTC:P0213	P0213:Cold Start Injector 1 Malfunction International standard protocol: http://www.obd-codes.com/trouble_codes/
883	DTC Remained	CFG:DTC:B0001:5	DTC:B0001 Means:DTC Code 5:Means: Continued time 5hr
998	external car alarm type	CFG:[external alarm type]	
999	External alarm report	CFG:[External alarm event]	

3-1. Store and forward function:

Forward: After the unit connects server, and the GSM signal is good, it will start to forward report saved in memory.

Packet of report: The unit will collect 1, 2 but max is 3 reports in 1 packet. Store forward packet includes 1,2 or 3-piece of reports. Each report has the header (\$EXT) and end byte ([CR][LF]).

Please refer to example below:

1. One packet with one reports:

\$EXT,GT3100,A,130401025807,N2247.5221E11402.7513,012,244,NA,NA,8,15,6571.46,NA,05020100,108

2. One packet with two reports

\$EXT,GT3100,A,130401025807,N2247.5221E11402.7513,012,244,NA,NA,8,15,6571.46,NA,05020100,108

\$EXT,GT3100,A,130401025907,N2247.5221E11402.7513,012,244,NA,NA,8,15,6571.46,NA,05000100,254

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	XA1	Door Lock	&&[ID],XA1[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],301[CR][LF]	Control the UNIT's door lock output		
	XA2	Door Unlock	&&[ID],XA2[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],302[CR][LF]	Control the UNIT's door unlock output		
	XA3	ARMING	&&[ID],XA3[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],303[CR][LF]	Control the UNIT's Arming output		
	XA4	Activate PANIC	&&[ID],XA4[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],304[CR][LF]	Control the UNIT's Lamp & Siren output, to alarming for 30 seconds.		
	XA5	Activate Anti-carjacking	&&[ID],XA5[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],305[CR][LF]	Control the UNIT go to Anti-carjacking mode		
	XA6	Emergency release	&&[ID],XA6[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],306[CR][LF]	Control the UNIT go to Disarming mode (to deactivate Arming, Panic and Anti-carjacking)		
	XA7	Trunk Release	&&[ID],XA7[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],307[CR][LF]	Control the UNIT's trunk output		
	XAA	Channel 2 output	&&[ID],XAA[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],310[CR][LF]	Control the UNIT's channel 2 output		
	Y01	Request current location	&&[ID],Y01[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],401 [CR][LF]	Request the UNIT to report location.		
	Y02	Wakeup the UNIT	&&[ID],Y02[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 402 [CR][LF]	Wakeup the UNIT (Note that this message will send to the UNIT by SMS only.)	Wake up the UNIT to go online / connect to GPRS and server. &&[ID],Y02[CR][LF] When connected, the unit will send a report \$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 402 [CR][LF] (Note that this message will send to the UNIT by SMS only.)	

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	Y03	Let the UNIT to go offline in GSM mode	&&[ID],Y03[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 403 [CR][LF]	Let the UNIT to go offline in GSM mode. (Note that this message will send to the UNIT via GPRS only.)	Let the UNIT to go offline in GSM mode. &&[ID],Y03[CR][LF] When disconnected, the unit will send a SMS report \$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 403 [CR][LF] (Note that this message will send to the UNIT by GPRS only.)	
	Y04	Let the system temporarily repor	&&[ID],Y04,[trace duration],[trace interval][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],404 [CR][LF]	Let the system temporarily report in [trace duration] and using the [report interval] to send the report	The UNIT will send report in interval defined by [trace interval] and will stop after duration defined by [trace duration].	
	Y05	Let unit don't send store forward data from the memory	&&[ID],Y05[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],405 [CR][LF]	Let unit don't send store forward data from the memory	The UNIT will stop sending report of store forward.	
	Y06	Read firmware version	&&[ID],Y06[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 406, CFG:Y06, [firmware version] [CR][LF]	Read firmware version (Message: CFG:Y06,firmware version)		
	Y07	Read Geo-fence Status	&&[ID],Y07[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 407, CFG:Y07, [Geo-fence status] [CR][LF]	Read Geo-fence Status (Message: CFG: Y07, CCRRCCRR :0000-1F1F)	Check the UNIT Geo-fence status. This command is to detect the UNIT is within which Geo-fence region. &&[ID],Y07[CR][LF] \$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 407, CFG:Y07, CCRR [CR][LF] Geo-fence status format :CCRR: Range :0000-1F1F The high byte CC represents the circular Geo-fence status and 'RR' represents the rectangular Geo-fence status. It is a HEX transformation of one byte, but only the last five digits is used to represent the Geo-fence trigger in the block '0' to '4'. '0000' means the unit is none of the preset Geo-fence event is triggered. (HEX transformation 00000000 --à 00, and 00000000--à 00) '0100' means the circular Geo-fence set '0' is triggered, but not in rectangular set. (00000001 --à01, 00000000 --à 00) '0012' means the rectangular Geo-fence set '1' and '4' are triggered, but nothing changes in circular Geo-fence (00000000 --à 00, 00010010--à 12)	

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	Y08	Read Battery storage level Status	&&[ID],Y08[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 408, CFG:Y08, [external battery level]] [CR][LF]	Read Battery storage level Status (Message: CFG: Y08, xxxx is the [battery level]: 01-99)	External battery level (xx): 01-99 The value is the percent of the battery capacity, We use ASCII characters to represent the status of battery, 99 means the storage is 99% of the battery capacity, if the battery level is lower than the BATTERY ALARM set from the PC setup program, it will generate Event “111”.	
	Y09	reset the GSM module	&&[ID],Y09[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 409 [CR][LF]	reset the GSM module	Unit will send back the report, and reset the GSM module, when received the command.	
	Y12	Set the preset distance	&&[ID],Y12,[distance] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 412 [CR][LF]	Set the preset distance (Message: CFG: Y12, preset distance)		
	Y20	Read the unit IMEI	&&[ID],Y20[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 420, CFG:Y20, [IMEI]] [CR][LF]	Read the unit IMEI	Read the unit IMEI (Message: CFG: Y20, IMEI.)	
	Y35	reset the GPS module	&&[ID],Y35[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 435 [CR][LF]	Unit will send back the report, and reset the GPS module, when received the command.		
	Y36	reset the unit	&&[ID],Y36[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 436 [CR][LF]	Unit will send back the report, and reset the unit, when received the command.	Send Y36 Command return 436,CFG:ACC ON (ACC ON reset the unit cannot be performed immediately, to wait until after the implementation of the ACC OFF reset the unit) Send Y36 Command return 436 (Implementation of reset the unit now.)	
	Y83	Control buzzer chirp	&&[ID],Y83[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 483 [CR][LF]	Control buzzer chirp	means buzzer will chirp 1sec when received Y83 command	
	Y96	Request external car alarm type	&&[device id],Y96[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],998,CFG:[Request external car alarm type] [CR][LF]	Event:998 Request external car alarm type		
	Y97	Exit OTA function	&&[ID], Y97 [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],497 [CR][LF]	Exit OTA function &&[ID],Y97[CR][LF] The UNIT will exit OTA function		

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	Z01	Setup dial-up interval, reconnect times and dial-up mode	&&[ID],Z01,[time interval],[retry times] , [connection mode][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 501, CFG: Z01, [time interval],[retry time], [connection mode] [CR][LF]	Setup the time interval (in seconds) and connection mode. &&[ID],Z01,[time interval], [retry time], [connection mode][CR][LF] [time interval]: reconnect interval time in seconds. [retry time]: The Max. reconnect times [connection mode]: 1 ,2, or 3, to select “GPRS always on-line”, “Base on report mode” or “GPRS connect once”	[time interval]: in seconds, e.g. 600 means 600 seconds. [retry times]: retry times to connect server while fail to connect to server. [connection mode]: 1,2 or 3, to select “GPRS always on-line”, “Base on report mode” or “GPRS connect once” “GPRS always on-line”: connection mode is “1”, While using this mode, when the unit can not searched GPRS signal, system will reconnect GPRS interval a preset value. (e.g.: 1minute) “Base on report mode”: connection mode is “2”, While using this mode, the unit will connect to the server when there is a report to send. If the first connection is failed, it will retry to connect to the server up to the max. reconnect times. Each retry will be separated by the reconnect “interval”. “GPRS connect once”: connection mode is 3. While using this mode, the unit will connect to the server when there is a report to send (but only try once). If it is not successful, the report will be stored and sent out in the next successful connection. Disconnect GPRS connection when report sending is completed. (This is suitable for GSM operator testing purpose.)	Example 1: &&[ID],Z01,600,0,1[CR][LF] means the UNIT work on “GPRS always on-line” mode, the UNIT connect to the server every 600 secs (10 minutes) when the unit can not searched GPRS signal, and it will “retry” for ever. Example 2: &&[ID],Z01,600,2,2[CR][LF] means the UNIT work on “Base on report mode”, the unit will connect to the server when there is a report to send. If the first connection is failed, it will retry to connect to the server up to 2 times. Each retry will be separated by the 600 secs (10 minutes). Example 3, &&[ID],Z01,0,0,3[CR][LF] means the UNIT work on “GPRS connect once” mode, the unit will connect to the server when there is a report to send (but only try once). If it is not successful, the report will be stored and sent out in the next successful connection. Disconnect GPRS connection when report sending is completed.
	Z02	Setup the Geo-fence area	&&[ID],Z02,[index],[origin],[radius],[scope][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 502, CFG: Z02, [index],[origin],[radius][scope] [CR][LF]	Description: Setup the Circular Geo-fence zone and send to the UNIT. When the UNIT receives this packet, it will store in the UNIT’s ram. The server will approximate the Geo-fence area into a circular shape, and send two values to the UNIT, (i) the original (latitude, longitude) and (ii) the radius (km). Syntax: &&[ID],Z02, [index] ,[origin],[radius],[scope] [CR][LF] [index]: Circular type Geo-fence locator 0 to 4. There will be up to 5 sets circular type Geo-fence area can be recorded in the UNIT’s ram. ‘INDEX’ = 4 will be used for Immediate Geo-fence use. (so no origin will be set.) [origin]: 21 digits standard format, e.g. N2446.5321E12120.4231 [radius]: in km, e.g. 1.88 means 1.88 km [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	[index]: 0 to 9. Circular type Geo-fence locator [origin]: 21 digits standard format, e.g. N2446.5321E12120.4231 [radius]: in km, e.g. 1.8 means 1.8 km [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	Example 1, &&[ID],Z02,0,N2446.5321E12120.4231,01.88,1[CR][LF] Represents the Geo-fence area at index ‘0’ is originated at “North 24 degrees 46.5321 minutes, East 121 degrees 20.4231 minutes” and the radius is 1.88km. Example 2, to disable/enable the in-car Geo-fence index ‘1’ without change the previous setting: ‘&&[ID],Z02,,,0[CR][LF]’ and ‘&&[ID],Z02,0,,,1[CR][LF]’ Example 3, &&[ID],Z02,0,?,?,?[CR][LF] means the server is requesting the UNIT to send back the current settings in ram index‘0’. Example 4: &&[ID],Z02,4,,1[CR][LF] Setup the Immediate Geo-fence radius to 1 km.

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	Z02	Setup Immediate Geo-fence radius (Circular only)	&&[ID],Z02,4,,[radius],[scope][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 502, CFG: Z02, 4,,[radius],[scope] [CR][LF]	Setup Immediate Geo-fence radius (Circular only) Event: 502 Message: CFG: Z02, 4,,[radius], [scope]	[radius]: in km, e.g. 1.8 means 1.8 km [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	
	Z03	Setup the Geo-fence area	&&[ID],Z03,[index],[point1],[point2],[scope][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 503, CFG: Z03, [index],[point1],[point2],[scope] [CR][LF]	Server will approximate the Geo-fence area into a rectangular shape, and will send two diagonal points/ locations into the UNIT. These two points will use the same (latitude, longitude) format. Syntax: &&[ID],Z03,[index], [point 1],[point 2],[scope][CR][LF] [index]: 0 to 4. [point1] and [point2]: diagonal points of the rectangular area, in 21 digits standard format, e.g. N2446.5321E12120.4231 [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	[index]: 0 to 4. Rectangular type Geo-fence locator [point1] and [point2]: diagonal points of the rectangular area, in 21 digits standard format, e.g. N2446.5321E12120.4231 [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	Example 1, &&[ID],Z03,0,N2446.5321E12120.4231,N2445.5321E12119.4231,1[CR][LF] Represents the Geo-fence area '1' is defined by these two diagonal points, e.g. point 1 is "North 24 degrees 46.5321 minutes, East 121 degrees 20.4231 minutes" and the point 2 is "North 24 degrees 45.5321 minutes, East 121 degrees 19.4231 minutes" Example 2, To enable or disable the in-car 'rectangular' Geo-fence '1' without changing the previous settings will need to use this command: &&[ID],Z03,1,,1[CR][LF] and &C03,1,,0[CR][LF] Example 3, &&[ID],Z03,1,?,?,?[CR][LF] means the server is requesting the UNIT to send back the index '1' settings.
	Z04	Ask for history repot by certain index range	&&[ID],Z04,[start],[end][CR][LF]	\$EXT,GT3100,A,130401025807,N2247.5221E114 02.7513,012,244,C30,NA,8,15,6571.46,NA,05020 100,510[CR][LF] \$EXT,GT3100,A,130401025807,N2247.5221E114 02.7513,012,244,C30,NA,8,15,6571.46,NA,05020 100,502[CR][LF] \$EXT,GT3100,A,130401025907,N2247.5221E114 02.7513,012,244,C30,NA,8,15,6571.46,NA,05020 100,501[CR][LF]	The server can send command to ask for history report data from the UNIT. Every 3 reports will be sent at a time.	[start index] :the report data order, the first one should be send 0~900 [end index]: the report data order, the last one should be send. 0~900 [start] must large than [end], that is, the "start" report is older than the end report. The older report will send out first. There are 3 reports to be out in every packet. e.g. &&[ID],Z04,510,501[CR][LF] 3 reports will be out in one packet.	&&[ID],Z04,510,501[CR][LF] The unit will send out ten records and their order is from 501 to 510. And the old data (510) report will send first, then followed by 509, 508 etc.
	Z05	Setup SMS or dial out telephone number	&&[ID],Z05,[index],[telephone number][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 505, CFG: Z05, [index],[telephone number] [CR][LF]	The server can send command to set the SMS or dial-out phone numbers.	[index]: 0 to 3 is index of SMS number, "0" means set primary SMS number. [telephone number]: telephone number used for SMS receive or dial out directly. e.g. &&[ID],Z05,0,+8613798428186[CR][LF]	&&[ID],Z05,0,+8613798413850[CR][LF] The unit will set the primary SMS number to +8613798413850.

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	Z08	Setup sleep mode parameters	&&[ID],Z08,[GSM power cut or not],[wake up time],[scope][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 508, CFG: Z08, [GSM power cut or not],[GPS power cut or not], [wake up time] ,[scope] [CR][LF]	The server can send command to set the Sleep MODE parameters.	[GSM power cut or not]: 0 or 1,"1" means cut GSM power while in sleep mode, "0" never cut GSM power in sleep mode. [GPS power cut or not]: 0 or 1,"1" means cut GPS power while in sleep mode, "0" never cut GPS power in sleep mode. [wake up time]: set unit wake up wait time in minutes. e.g. &&[ID],Z08,1,1,120 Unit will sleep 120 minutes then auto wakeup once to search GPS and send a wake up report. In sleep mode cut power both GPS and GSM.	&&[ID],Z08,1,1,120 Unit will sleep 120 minutes then auto wakeup once to search GPS and send a wake up report. In sleep mode cut power both GPS and GSM
	Z09	Setup Point Geo-fence	&&[ID],Z09,[index],[point],[scope][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 509, CFG: Z09,[index],[point],[scope] [CR][LF]	The server can send command to set the point Geo-fences.	[index]: 0 to 19 index of point [point]: longitude and latitude of point. It must use our standard format. [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms e.g. &&[ID],Z09,0,N2240.1234E11359.1234,1 [CR][LF]	e.g. &&[ID],Z09,0,N2240.1234E11359.1234,1 [CR][LF] Set index "0" point Geo-fence to be at N2240.1234E11359.1234. &&[ID],Z09,0,0[CR][LF] Disable first point Geo-fence.
	Z10	Setup APN , user name, password	&&[ID],Z10,[APN],[user name],[password][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 510,CFG: Z10, [APN],[user name],[password] [CR][LF]	The server can send command to set the APN, user name, password	[APN] GPRS network name [user name] [password]	&&[ID],Z10,?,?,?[CR][LF] check APN , user name and password. e.g. &&[ID],Z10,cmnet,,[CR][LF] [APN]: cmnet means china mobile net
	Z11	Setup Reduce dial-up procedure parameters	&&[ID],Z11,[Maximum Reconnect times],[Idle time],[on/off][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 511,CFG: Z11, [maximum reconnect time],[Idle time],[on/off] [CR][LF]	The server can send command to set the dial up procedure	if the "reduce dial-up procedure" is on, the dial-up times will be limited by the [maximum reconnect times] if procedure continues failed. When the maximum reconnect times is achieved, the unit will be forced to wait for [idle time] before the next reconnection. [Maximum Reconnect times]: total dial-up retry times [Idle time]: defined in sec. The UNIT will wait for [Idle time] before next dial-up retry. For example, [Idle]=3600, means 3600 seconds. [on/off]: 0 or 1, turn on / off reduce dial-up procedure.	&&[ID],Z11,10,3600,1[CR][LF] After retry dial-up 10 times, the UNIT will wait for 3600 seconds before make any new dial-up connection.
	Z12	Setup the time interval for the 'keep alive' procedure	&&[ID],Z12,[time interval],[retry times],[on/off][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 512, CFG: Z12, [time interval],[retry time], [on/off] [CR][LF]	Setup the time interval (in seconds) for the 'keep alive procedure'. &&[ID],Z12,[time interval], [retry time], [on/off][CR][LF] [time interval]: seconds. [retry time]: how many times the keep alive package will be resent. [on/off]: 1 or 0, to enable this procedure or not	[time interval]: in seconds, e.g. 600 means 600 seconds. [retry times] times for the keep alive package will be resent. [on/off]: 0 or 1 to enable this procedure or not	Example 1: &&[ID],Z12,600,2,1[CR][LF] means the UNIT will send keep alive message to the server every 600 secs (10 minutes) and it will retry 2 time if the first packet is not received successfully. Example 2: &&[ID],Z12,600,2,0[CR][LF] means no keep alive procedure. Example 3, &&[ID],Z12,?,?,?[CR][LF] means the server is requesting the UNIT to send back the current settings.

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	Z13	Setup the unit device ID and device password	&&[ID],Z13,[device id],[device password] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],513,CFG:Z13,[device id],[device password], [CR][LF]	Setup the unit device ID and device password,	[device id]: in digit [device password]: in digit (this parameter is not necessary.)	&&[ID],Z13,GT3000M,1234,[CR][LF] The unit device ID is GT3000M, and password is 1234
	Z14	Setup GPS fix mode	&&[ID],Z14,[fix mode][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],514,CFG: Z14,[fix mode] [CR][LF]	The server can send command to setup the GPS fix mode.	[fix mode]: 0 ---- Auto mode 1 ---- 3D mode 2 ---- 2D&3D mode	Example: &&[ID],Z14,1[CR][LF] Setup the GPS in 3D mode
	Z17	Setup TL-sensor trigger degree & report	&&[ID],Z17,[angle degree],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],517,CFG: Z17,[angle degree],[scope] [CR][LF]	Setup TL-sensor trigger degree & report	[angle degree]: 3~20 degree [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	&&[ID],Z17, 5,1 [CR][LF] The TL-sensor angle > 5 degree will be trigger & send trigger report in local.
	Z21	Setup report after keep alive ACK	&&[ID],Z21,[on/off][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 521,CFG: Z21,[on/off] [CR][LF]	If you active this function, report will be sending only after received the keep alive ACK. So GPRS packet lost will be prevent.	[on/off]: 0 or 1, sending report after keep alive ACK or not.	&&[ID],Z21,1[CR][LF] Enable report after keep alive ACK
	Z22	Setup fake connection prevent	&&[ID],Z22,[idle time],[on/off][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 522,CFG: Z22,[idle time],[on/off] [CR][LF]	If the GPRS network is idle(no data send/receive) for a long time, then the unit will send a keep alive packet to check if the connection is fake before any reports.	[idle time]: in seconds. The idle time to prevent fake connection [on/off]: 0 or 1. Enable or disable the fake connection prevent function.	&&[ID],Z22,1800,1[CR][LF] If the GPRS network is idle for 30Min(1800Sec), then send a keep-alive packet to check if the connection is fake before any reports.
	Z25	Setup FN700 send Add fuel or Steal fuel report	&&[ID],Z25,[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 525, CFG: Z25, [scope][CR][LF]	Setup FN700 send Add fuel or Steal fuel report Event: 525 Message: CFG:Z25, [scope]	[scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	
	Z27	Setup lock GPS when ACC off	&&[ID],Z27,[avoid towed distance],[Lock GPS on/off] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 527, CFG: Z27, [avoid towed distance],[Lock GPS on/off] [CR][LF]	Setup lock GPS when ACC off Event: 527 Message: CFG: Z27,[avoid towed distance],[Lock GPS on/off]	[avoid towed distance]: 0.5 - 300 km, if system check GPS distance is more this value, then GPS will unlock. [Lock GPS on/off]: 1 or 0. Enable or disable lock GPS function, if enable lock GPS, then system will lock GPS (the reports speed is 0, location is same), when ACC off.	&&[ID],Z27, 1, 1 [CR][LF]
	Z29	Setup low fuel report	&&[ID],Z29,[Fuel level],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],529,CFG:Z29,[Fuel level],[scope] [CR][LF]	Setup low fuel report Event: 529 Message: CFG: Z29,[Fuel level],[scope]	[Fuel level]: 00 to 99, e.g. if =10, it will generate a low fuel warning report. [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	&&[ID], Z29, 50, 1 [CR][LF] Means the low fuel warning report is enabled and the UNIT will send report to the server in local if the fuel level is less than 50%.

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	Z31	Setup Fixed time report	&&[ID], Z31,[time move local],[time stop roaming],[time stop roaming],[time move sms],[time stop sms],[threshold speed],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 531, CFG: Z31,[time move local],[time stop local],[time move roaming],[time stop roaming],[time move sms],[time stop sms],[threshold speed],[scope] [CR][LF]	Setup Fixed time report Event 531 Message: CFG: Z31,[time move local],[time stop local],[time move roaming],[time stop roaming],[time move sms],[time stop sms],[threshold speed],[scope]	[time move local]: in seconds, e.g. 600 means 600 sec., 10 means 10 seconds. No decimal. [time stop local]: in seconds, e.g. 600 means 600 sec., 10 means 10 seconds. No decimal. [time move roaming]: in seconds, e.g. 600 means 600 sec., 10 means 10 seconds. No decimal. [time stop roaming]: in seconds, e.g. 600 means 600 sec., 10 means 10 seconds. No decimal. [time move sms]: in seconds, e.g. 600 means 600 sec., 10 means 10 seconds. No decimal. [time stop sms]: in seconds, e.g. 600 means 600 sec., 10 means 10 seconds. No decimal. [threshold speed]: debounce speed to determine the vehicle is moving or not (km/h). Typical value 6 km/h. Note: if set it to 0 then means unit pass ACC on/off to determine the vehicle is moving or not. [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	
	Z33	Setup Intelligent history	&&[ID],Z33,[moving interval],[distance],[stop interval],[on/off][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 533, CFG: Z33, [moving interval],[speed],[stop interval],[on/off] [CR][LF]	Similar to the above report method, this setting will tell the UNIT to record the report in the UNIT's ram.	[moving interval]: Report time interval---No more than (moving) in (sec). [distance]: moving distance(km) [stop interval]: Report time interval---No less than (stopped) in sec. [on/off]: 1 or 0, to enable this report or not.	Example 1, &&[ID],Z33,60,6,3000,1[CR][LF] means the UNIT will record the full report message to the UNIT's ram every 3000 sec., when the vehicle's speed over 6 km/hr, it will record it to the UNIT's ram every 60sec. Example 2, &&[ID],Z33,?,?,?[CR][LF] means the server is requesting the UNIT to send back the current settings.
	Z34	Setup the Maximum speed limit	&&[ID],Z34, [max. speed],[enter delay time],[exit delay time],[scope],[resend time],[alert on/off] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 534, CFG: Z34, [max. speed],[enter delay time],[exit delay time],[scope],[resend time],[alert on/off] [CR][LF]	If the vehicle goes beyond preset speed limit, it will generate a report.	[max. speed]: in km/hour [enter delay time]: enter speed over debounce time [exit delay time]: exit speed over debounce time [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms [resend time]: in seconds. Speed over report resend interval. [alert on/off]: 1 or 0. to enable/disable alert when speed over. (Note: this is optional parameter. It is just for the products with buzzer. For some products without buzzer, this parameter is not necessary.)	&&[ID],Z34,80,30,20,1,60,1[CR][LF] If speed over 80km/h for 30sec then speed over. If speed low than 80km/h for 60sec then speed recover. Send speed over report every 60 sec, and buzzer alert when speed over. Speed over report can be sent in case of local.
	Z36	Setup the fixed distance report	&&[ID],Z36,[distance local],[distance roaming],[distance sms],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 536, CFG: Z36,[distance local],[distance roaming],[distance sms],[scope] [CR][LF]	When the vehicle moves 'x' km, the UNIT will send an event to the server. Setup the fixed distance report Event 536 Message: CFG: Z36,[distance local],[distance roaming],[distance sms],[scope]	[distance local]: 0-300 moving distance(km) [distance roaming]: 0-300 moving distance(km) [distance sms]: 0-300 moving distance(km) [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	&&[ID],Z36,1,1,1,1[CR][LF] It means the unit will send distance report to the server every 1km, and report can be sent in case of local.

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	Z37	Setup Course change report	&&[ID],Z37,[course change in degree],[debounce time],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 537, CFG: Z37, [course change in degree],[debounce time],[scope] [CR][LF]	Setup Course change report Event 537 Message: CFG: Z37, [course change in degree],[debounce time],[scope]	[course change in degree]: 000-360 in degree, no decimal. [debounce time]: course change debounce time, in second [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	&&[ID],Z37,180,10,1[CR][LF] It means the course change report will be sent in local when the vehicle change the direction more than 180 degree for 10 sec.
	Z38	Setup Temperature report	&&[ID],Z38,[minimum Temp],[maximum Temp],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 538, CFG: Z38,[minimum Temp],[maximum Temp],[scope] [CR][LF]	Setup Temperature report Event 538 Message: CFG: Z38, [min Temperature] ,[max Temperature], [scope]	[minimum temp] & [maximum temp] are the temperature can be in Celsius. The formats are C-8, C0.... (no decimal.) [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms Note: if the [scope] > 0 then the AD port will be auto changed to temperature function.	It means the temperature report will be sent in local if the temperature is not in the range between -5 °C and +10 °C. Example 2: &&[ID],Z38, C-5,C40,0[CR][LF] It means the temperature setting is between -5 °C and 40 °C, but this report mode is disabled. Example 3, &&[ID],Z38,?,?,?[CR][LF] It means the server is requesting the UNIT to send back the current settings.
	Z39	Remote setup server IP and port	&&[ID],Z39,[sck type],[IP Address],[port],[on/off] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 539, CFG: Z39, [socket type],[IP Address],[port], [on/off] [CR][LF]	Event 539 Message: CFG: Z39, [socket type],[IP Address],[port],[on/off]	[sck type] socket type ,0 or 1, 0 is UPD ,1 is TCP. [IP Address] new IP address of server, e.g. 192.168.1.138 [port] new IP port, e.g. 3000 [on/off] 1 or 0.set this configuration active or de-active	Example1 :&&[ID],Z39,1,192.168.1.138,3000,1[CR][LF] means set let the unit connect to "192.168.1.138",the port is 3000. and use TCP protocol. Example 2, &&[ID],Z39,0,?,?,?[CR][LF] means the server is requesting the UNIT to send back the current UDP server settings. Example 3, &&[ID],Z39,1,?,?,?[CR][LF] means the server is requesting the UNIT to send back the current TCP server settings.
	Z40	Setup Low Battery report	&&[ID],Z40,[lowest battery level for alarm],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 540, CFG: Z40, [lowest battery level for alarm], [scope] [CR][LF]	Event 540 Message: CFG: Z40, [lowest battery level for alarm], [scope]	[lowest battery level for alarm]: 00 to 99, e.g. if =10, it will generate a low battery warning report. [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	Example1: &&[ID],Z40, 30,1[CR][LF] It means the low battery warning report will be sent to the server in local if the battery level is less than 30%. Example 2, &&[ID],Z40,?,?[CR][LF] It means the server is requesting the UNIT to send back the current: low Battery report settings.
	Z43	Setup crash report	&&[ID],Z43,[crash threshold],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 543, CFG: Z43, [crash threshold], [scope] [CR][LF]	Event 543 Message: CFG: Z43, [crash threshold], [scope]	[crash threshold]: 20~160(i.e. 2~16G) [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	Example1:&&[ID],Z43,50,3[CR][LF] It means the unit will send a crash report when acceleration over 5 G,please note, numeric 50 means acceleration is 5G Example2:&&[ID],Z43,?,?[CR][LF] It means the server is requesting the UNIT to send back the crash settings.
	Z44	Setup stop report	&&[ID],Z44,[scope],[threshold speed],[debounce time] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 544, CFG: Z44, [scope],[threshold speed],[debounce time] [CR][LF]	Event 544 Message: CFG: Z44, [scope],[threshold speed],[debounce time]	[scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms [threshold speed]: in km/hour [debounce time]: in second	&&[ID],Z44,1,5,20[CR][LF] It means the unit will send a stop report in local if unit detected speed is less than 5km/h and lasts 20 seconds.

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	Z47	Setup id-card report	&&[ID],Z47,[scope][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 547, CFG: Z47, [scope] [CR][LF]	Event 547 Message: CFG: Z47, [scope]	[scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	&&[ID],Z47,1[CR][LF] It means the UNIT will send id-card report to the server in local when system checked ID Card
	Z48	Setup id-card immobilization functions	&&[ID],Z48,[E-card control startkill enable/disable][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],548, CFG: Z48, [enable/disable] [CR][LF]	Event 548 Message: CFG: Z48, [enable/disable]	[enable/disable]: 1 or 0. to enable immobilization functions or not.	Example: &&[ID],Z48,1[CR][LF] means enable immobilization functions, the engine will be immobilized after ACC off 2 minutes, using id-card can release engine.
	Z49	Setup using id-card or not	&&[ID],Z49,[enable/disable][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],549, CFG: Z49, [enable/disable] [CR][LF]	Event 549 Message: CFG: Z49, [enable/disable]	[enable/disable]: 1 or 0. to enable the functions of id-card or not	Example: &&[ID],Z49,1[CR][LF] means the functions of E-card will be used. The functions about sending id-card report and engine immobilization can be used.
	Z73	Set steal fuel report(FU700)	&&[ID],Z73,[scope],[decrease percent][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 573, CFG: Z73,[scope],[decrease percent] [CR][LF]	Set steal fuel report Event: 573 Message: CFG: Z73,[scope],[decrease percent]	[scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms [decrease percent]: percent value of fuel is decreased, such as, 5%	
	Z74	Set add fuel report(FU700)	&&[ID],Z74,[scope],[increase percent][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status], 574, CFG: Z74,[scope],[increase percent] [CR][LF]	Set add fuel report Event: 574 Message: CFG: Z74,[scope],[increase percent]	[scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms [increase percent]: percent value of fuel is increased, such as, 10%	
	Z75	Setup ID card parameters	&&[ID],Z75,[index],[card id][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],575,CFG:[index],[card id][CR][LF]	Event 575 Message: CFG: Z75,[index],[card id][CR][LF]	[index]: 0--19 0 means set the first ID card; 1 means set the second ID card; 2 means set the third ID card..... [card id]: number of ID card;	
	Z79	Report for INPUT1 (In-vehicle DOOR) changes Enable or Disable	&&[ID],Z79,[scope][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],579,CFG: Z79,[scope] [CR][LF]	Report for INPUT1 (In-vehicle DOOR) changes Enable or Disable Event 579	[scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	
	Z80	Report for In-vehicle ACC changes Enable or Disable & ACC off report	&&[ID],Z80,[Acc trigger scope],[self-geofence on/off][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],580,CFG: Z80, [Acc trigger scope],[self-geofence on/off] [CR][LF]	Report for In-vehicle ACC changes Enable or Disable Event 580	[Acc trigger scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms [self-geofence on/off]: 0---disable 1---enable Automatic open self-geofence function when ACC OFF	

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	Z98	Send the software information to the unit	&&[ID],Z98,[socket type],[IP address],[port],[firm version],[upgrade indication] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],598, CFG: Z98,[socket type],[IP address],[port],[firm version],[upgrade indication] [CR][LF]	Message: CFG: Z98,[socket type],[IP address],[port],[firm version],[upgrade indication] [CR][LF] Unit will reply report that event code is 598 to server when the unit received Z98 command from the server. Later, the unit will cut current GPRS connection to connect upgrade server when unit accept air download command. Unit will reset by self and reconnect to the server after upgrade software is finished. Unit will send report that event code is 200 to the server when upgrade software was successful. Unit will send report that event code is 199 to the server when upgrade software was failed.	[socket type]: the software upgrade server socket type ,0 or 1, 0 is UPD ,1 is TCP. [IP Address]: IP address of software upgrade server, e.g. 192.168.1.138 [port]: software upgrade server IP port, e.g. 3000 [software version]: the software version that will be upgraded,e.g.GT3000_GT3000B.B_PTM_5.1.0_1.05 [upgrade indication]: There is 1/2/3 /5/6/7 in the command When SERVER Send to Unit: 1: the current version is older than the command give version will be upgraded when ACC off 2: the current version is newer than the command give version will be upgraded when ACC off 3: upgrade unconditionally when ACC off 5: the current version is older than the command give version will be upgraded 6: the current version is newer than the command give version will be upgraded 7: upgrade unconditionally	There is a/b/c/d/e in the report When UNIT Reply to Server: 1 -- up grade accept, and will start immediately 2 -- upgrade accept, and will start when ACC OFF 3 -- upgrade not accept, server IP address invalid 4 -- upgrade not accept, product name not matched 5 -- upgrade not accept, PCB version invalid 6 -- upgrade not accept, firmware version invalid 7 -- upgrade not accept, other errors
	ZE13	Set fuel sensor ID number (FN700)	&&[ID],[Password], ZE13,[index],[fuel sensor ID number] ,[fuel height increased],[fuel height reduced][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],732, CFG: ZE13,[index],[fuel sensor ID number],[fuel height increased],[fuel height reduced] ,[fuel sensor firmware version] [CR][LF]	Event: 713 Message: CFG: ZE13,[index],[fuel sensor ID number],[fuel height increased],[fuel height reduced],[fuel sensor firmware version]	[[index]: 0----- fuel sensor 1; 1-----fuel sensor 2; 2----- fuel sensor 3; [fuel sensor ID number]: fuel sensor ID number [fuel height increased]:450(Unit :1/10MM) [fuel height reduced]:450(Unit :1/10MM)	
	ZE32	Setup fuel sensor type	&&[ID],ZE32, [scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],732, CFG: ZE32,[scope] [CR][LF]	Setup fuel sensor type Event 732	[scope]:0----FU700 1----FN700	
	ZE33	Setup wired or wireless temperature sensor	&&[ID],ZE33, [scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],733, CFG: ZE33,[scope] [CR][LF]	Setup wired or wireless temperature sensor Event 733	[scope]:0----wired temperature sensor 1----wireless tmeperature sensor	
	ZE34	Setup wireless door status	&&[ID],ZE34, [scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],734, CFG: ZE34,[scope] [CR][LF]	Setup wireless door status Event 734	[scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	ZE45	Setup trailer report	&&[ID], ZE45,[scope],Trailer distance][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],745,CFG: ZE45,[scope],[avoid towed distance][CR][LF]	Setup trailer report Event 745	[scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms [trailer distance]: 0.5 - 300 km	&&[ID],ZE45,1,3 [CR][LF] After the ACC off, the unit will lock GPS, when car be towed (the GPS distance is more than 1km,) send towed report (the towed report event code is 138)
	ZE51	Setup turn-over report	&&[ID], ZE51,[turn over degree],[scope][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],751,CFG: ZE51,[turn over degree],[scope][CR][LF]	Send turn-over report when angle over the setting value. Event 751	[turn-over degree]: 75~180 degree [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	&&[ID],ZE51,90,3 [CR][LF] It means the unit will send a turn-over report when angle over 90 degree. Example2:&&[ID],ZE51,?,?[CR][LF] It means the server is requesting the UNIT to send back the turn-over settings.
	ZE52	Setup drift report	&&[ID], ZE52,[drift acceleration],[scope][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],752,CFG: ZE52,[drift acceleration],[scope][CR][LF]	Send vehicle-drift report when acceleration over the setting value. Event 752	[drift acceleration]: 20~160(i.e. 2~16G) [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	&&[ID],ZE52,40,3 [CR][LF] It means the unit will send a drift report when acceleration over 4 G, please note, numeric 40 means acceleration is 4G Example2:&&[ID],ZE52,?,?[CR][LF] It means the server is requesting the UNIT to send back the drift settings.
	C04	Setup recording CANBUS/OBD report	&&[ID],C04,[index],[enable or not],[record interval] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3004,CFG:C04,[index],[enable or not],[record interval] [CR][LF]	Event 3004 Message: CFG: C04,[index],[enable or not],[record interval]	[index]: 0 --- set T1 report 1 --- set T2 report 2 --- set T3 report 3 --- set T4 report 4 --- set T5 report [enable or not]: 0 --- disable; 1 --- enable; [record interval]: recording interval of time in second	
	C05	Setup sending CANBUS/OBD report	&&[ID],C05,[index],[scope],[send interval][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status]3005,CFG:C05,[index],[scope],[send interval] [CR][LF]	Event 3005 Message: CFG: C05,[index],[scope],[send interval]	[index]: 0 --- set T1 report 1 --- set T2 report 2 --- set T3 report 3 --- set T4 report 4 --- set T5 report [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms [send interval]: sending interval of time in second	
	C07	Setup RPM over running report	&&[ID],C07,[RPM threshold],[out RPM threshold time],[scope][CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3007,CFG:C07,[RPM threshold],[out RPM threshold time],[scope] [CR][LF]	Event 3007 Message: CFG: C07,[RPM threshold],[out RPM threshold time],[scope]	[RPM threshold]: such as, 5000 [out RPM threshold time]: out RPM threshold time in second, such as 5 seconds [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	C08	Setup Abnormal Fuel decrease report	&&[ID],C08,[FUEL lost],[distance],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3008,CFG:C08,[FUEL lost],[distance],[scope] [CR][LF]	Event 3008 Message: CFG: C08,[FUEL lost],[distance],[scope]	[FUEL lost]: such as, 5% [distance]: such as 1 km [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	
	C09	Setup Fuel consumption report	&&[ID],C09,[average fuel consumption],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3009,CFG:C09,[average fuel consumption],[scope] [CR][LF]	Event 3009 Message: CFG: C09,[average fuel consumption],[scope]	[average fuel consumption]: such as, 7 L/100km [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	
	C10	Setup Engine idle time report	&&[ID],C10,[Engine idle time],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3010,CFG:C10,[Engine idle time],[scope] [CR][LF]	Event 3010 Message: CFG: C10,[Engine idle time],[scope]	[Engine idle time]: such as, 5 minutes [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	
	C11	Setup Engine over heat report	&&[ID],C11,[Engine temperature threshold],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3011,CFG:C11,[Engine temperature threshold],[scope] [CR][LF]	Event 3011 Message: CFG: C11,[Engine temperature threshold],[scope]	[Engine temperature threshold]: such as, 150 degrees Celsius [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	
	C13	Setup Over pressing fuel pedal report	&&[ID],C13,[press fuel pedal threshold],[press time],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3013,CFG:C13,[press fuel pedal threshold],[press time],[scope] [CR][LF]	Event 3013 Message: CFG: C13,[press fuel pedal threshold],[press time],[scope]	[press fuel pedal threshold]: such as, 95% [press time]: such as, 5 seconds [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	
	C14	Setup Emergency pressing brake pedal report	&&[ID],C14,[press brake pedal threshold],[press time],[speed threshold],[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3014,CFG:C14,[press brake pedal threshold],[press time],[speed threshold],[scope] [CR][LF]	Event 3014 Message: CFG: C14,[press brake pedal threshold],[press time],[speed threshold],[scope]	[press brake pedal threshold]: such as, 95% [press time]: such as, 5 seconds [speed threshold]: such as, 50 km/h [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	
	C15	Setup Driver overtime notice report	&&[ID],C15,[drive time],[scope],[exit delay time] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3015,CFG:C15,[drive time],[scope],[exit delay time] [CR][LF]	Event 3015 Message: CFG: C15,[drive time],[scope],[exit delay time]	[drive time]: it is the threshold time of driver overtime, such as, 8 hours [scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms [exit delay time]: after ACC off for this time such as 15 minutes, system will finish the count of drive time	
	C16	Setup Malfunction indicator lamp on	&&[ID],C16,[scope],[resend interval time] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3016,CFG:C16,[scope],[resend interval time] [CR][LF]	Event 3016 Message: CFG: C16,[scope],[resend interval time]	[scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms [resend interval time]: in hr	

Portman GPRS tracking command list

No	command	Command	GPRS Syntax	Reply to Server	description	Parameters:	Example
	C17	Setup duty on/off report	&&[ID],C17,[scope] [CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3017,CFG:C17,[scope] [CR][LF]	Event 3017 Message: CFG: C17,[scope]	[scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	
	C20	Reset OBD50/CAN50	&&[ID],C20[CR][LF]	\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],[mileage],[Id card number],[Status],3020[CR][LF]	Event 3020	[scope]: 0 --- disable; 1 --- enable local; 3 --- enable local + roaming; 5 --- enable local + sms; 7 --- enable local + roaming + sms	

Portman GPRS tracking Canbus & OBD

1. CANBUS/OBD regular reports

The reports of CANBUS/OBD include that as below:

‘\$BUSV1’ is packet header

[CR][LF] is packet tail

For example:

\$BUSV1,[ID],[T1],[date + time],[engine RPM],[speed],[Fuel consumption l/100km],[fuel consumption l/h],[engine temperature],[fuel level %],[battery voltage],[axleweight1 kg],[axleweight2 kg],[axleweight3 kg],[axleweight4 kg],[oil pressure],[ignition pedal],[torque],[event][CR][LF]

[ID]: The UNIT ID in alphanumeric string, which can be as long as 15 characters, such as: PORTMAN00001, or 1234567890...

[T1]: means No1 CANBUS/OBD report.

The details of CANBUS/OBD report as below:

a) T1 report

It is the fix time report of CANBUS/OBD information, T1 report will be defined as below:

Record CANBUS/OBD info every __ seconds, and send to server every __ seconds

Default: 10 sec, 60 sec

Report contents	Explanation
\$BUSV1,	
[ID],	
[T1],	
[date + time],	
[engine RPM R/M],	
[speed km/h],	

[Fuel consumption L/100km],	Current Fuel consumption L/100km
[Fuel consumption L/h],	Current Fuel consumption L/h
[engine temperature °C],	Current Engine temperature
[fuel level %],	Current Fuel remained
[battery voltage V],	Current voltage of battery
[axleweight1 kPa],	Current Bellow Pressure Front Axle Left (OBD not supported)
[axleweight2 kPa],	Current Bellow Pressure Front Axle Right(OBD not supported)
[axleweight3 kPa],	Current Bellow Pressure Rear Axle Left(OBD not supported)
[axleweight4 kPa],	Current Bellow Pressure Rear Axle Right(OBD not supported)
[oil pressure kPa],	Current Engine Oil Pressure
[acc pedal %],	Current Accelerator Pedal Position percentage
[torque %max],	Current Actual Engine - Percent Torque(OBD not supported)
[event]	
<CR><LF>	

a) T2 report

It is the last trip report of CANBUS/OBD information while ACC off, T2 report will be defined as below:
Record last trip CANBUS/OBD info while ACC off, and send to server while ACC off

Report contents	Explanation
\$BUSV1,	
[ID],	
[T2],	
[date + time],	
[max speed km/h],	Max speed of last trip
[max engine temperature °C],	Max Engine temperature of last trip
[max RPM R/M],	Max Engine rotation rate of last trip
[Average Speed km/h],	Average speed of last trip
[Average Fuel consumption L/100km],	Average Fuel consumption of last trip
[Trip Kilometers km],	Trip kilometers of last trip
[Trip Fuel used L],	Fuel used of last trip

[Trip Elapsed Time min],	Engine on time of last trip
[Trip Idle Time min],	Idle time of last trip
[rpm>threshold rpm sec],	time of RPM higher than rpm threshold setting in last trip
[speed>threshold speed sec],	time of speed higher than speed threshold setting in last trip
[PTO time sec],	total time for Power Take Out(PTO) in last trip(OBD not supported)
[cruise time sec],	total time of cruise in last trip(OBD not supported)
[brake apps times],	driver press break times in last trip(OBD not supported)
[clutch apps times],	driver press clutch times in last trip(OBD not supported)
[time torque>90% sec],	total time of torque more than 90% in last trip(OBD not supported)
[event]	
<CR><LF>	

a) T3 report

It is the total trip report of CANBUS/OBD information while ACC off, T3 report will be defined as below:

Record total trip CANBUS/OBD info while ACC off, and send to server while ACC off

Report contents	Explanation
\$BUSV1,	
[ID],	
[T3],	
[date + time],	
[Average Speed km/h],	Average speed of total trip
[Average Fuel consumption L/100km],	Average Fuel consumption of total trip
[Trip Kilometers km],	Trip Kilometers of total trip
[total fuel used liter],	fuel used of total trip
[total Elapsed Time min],	Engine on time of total trip
[total idle time min],	total idle time(engine started, but not move) of total trip
[rpm>threshold rpm sec],	time of RPM higher than rpm threshold setting in total trip
[speed>threshold speed sec],	time of speed higher than speed threshold setting in total trip
[total PTO time sec],	total time for Power Take Out(PTO) in total trip(OBD not supported)
[total cruise time sec],	total time of cruise in total trip(OBD not supported)

[brake apps times],	driver press break times in total trip(OBD not supported)
[clutch apps times],	driver press clutch times in total trip(OBD not supported)
[time torque>90% sec],	total time of torque more than 90% in total trip(OBD not supported)
[event]	
<CR><LF>	

a) T4 report

It is the fix time report of CANBUS/OBD information, T4 report will be defined as below:

Record CANBUS/OBD info every __ seconds, and send to server every __ seconds

Default: 10 sec, 60 sec

Report contents	Explanation
\$BUSV1,	
[ID],	
[T4],	
[date + time],	
[Fuel system 1 status],	bit0 --- 1 = Open loop - has not yet satisfied conditions to go closed loop bit1 --- 1 = Closed loop - using oxygen sensor(s) as feedback for fuel control bit2 --- 1 = Open loop due to driving conditions (e.g., power enrichment, deceleration enrichment) bit3 --- 1 = Open loop - due to detected system fault bit4 --- 1 = Closed loop, but fault with at least one oxygen sensor -may be using single oxygen sensor for fuel control bit5 --- 7 reserved (bits shall be reported as '0')
[calculated load value %],	Indicates a percentage of peak available torque. The normalized value of air mass per intake; stroke displayed as a percent.
[Fuel Rail Pressure KPa],	Indicates the fuel rail pressure at the engine referenced to atmosphere (gauge pressure)
[Intake Air Temperature °C],	Displays the intake manifold air temperature.
[Absolute Throttle Position %],	The absolute throttle position (not the relative or learned) throttle position. Usually above 0% at idle and less than 100% at full throttle.

[event]	
<CR><LF>	

a) T5 report (only CAN50)

It is the tachometer report of CANBUS/OBD information, T5 report will be defined as below:

Record tachometer CANBUS/OBD info while status changed, and send to server while status changed

Report contents	Explanation
\$BUSV1,	
[ID],	
[T5],	
[date + time],	
[Tachometer work state 1 0=rest 1=available 2=work 3=drive],	000 Rest - sleeping 001 Driver available – short break 010 Work – loading, unloading, working in an office 011 Drive – behind wheel 100-101 Reserved 110 Error 111 Not available
[Tachometer work state 2 0=rest 1=available 2=work 3=drive],	000 Rest - sleeping 001 Driver available – short break 010 Work – loading, unloading, working in an office 011 Drive – behind wheel 100-101 Reserved 110 Error 111 Not available
[Tachometer drive recognize 0=no 1=yes],	00 Vehicle motion not detected 01 Vehicle motion detected

[Tachometer time state 1 0=normal 1=15m before 4,5 hrs 2=4,5 hrs 3=15m before 9 hrs, 4=9hrs, 15 min before 16 hrs, 5=16hrs],	0000 Normal/No limits reached 0001 Limit #1 – 15 min before 4.5 h 0001 Limit #1 – 15 min before 4.5 h 0010 Limit #2 – 4.5 h reached 0010 Limit #2 – 4.5 h reached 0011 Limit #3 – 15 min before 9 h 0100 Limit #4 – 9 h reached 0101 Limit #5 – 15 min before 16 h (not having 8h rest during the last 24h) 0110 Limit #6 – 16 h reached 0111-1100 Reserved 1101 Other 1110 Error 1111 Not available
[Tachometer card 1 present 0=no 1=yes],	00 - Driver card not present 01 - Driver card present
[Tachometer over speed 0=no 1=yes],	Indicates whether the vehicle is exceeding the legal speed limit set in the ;tachograph. 00 No overspeed 01 Over speed

[Tachometer time state 2 0=normal 1=15m before 4,5 hrs 2=4,5 hrs 3=15m Before 9 hrs, 4=9hrs, 15 min before 16 hrs, 5=16hrs],	0000 Normal/No limits reached 0001 Limit #1 – 15 min before 4.5 h 0001 Limit #1 – 15 min before 4.5 h 0010 Limit #2 – 4.5 h reached 0010 Limit #2 – 4.5 h reached 0011 Limit #3 – 15 min before 9 h 0100 Limit #4 – 9 h reached 0101 Limit #5 – 15 min before 16 h (not having 8h rest during the last 24h) 0110 Limit #6 – 16 h reached 0111-1100 Reserved 1101 Other 1110 Error 1111 Not available
[Tachometer card 2 present 0=no 1=yes],	00 - Driver card not present 01 - Driver card present
[Tachometer event 0=no event 1=event],	Indicates that a tachograph event has occurred. This may include power supply interruption, interruption of the speed sensor, incorrect data on the driver card, driving without a driver card, illegal removal of a driver card, insertion of a driver card during driving, and time adjustment. 00 - No tachograph event 01 - Tachograph event
[Tachometer handling info 0=no 1=yes],	Indicates that handling information is present. Information could include “no; printer paper”, “no driver card”, etc. 00 - No handling information 01 - Handling information

[Tachometer performance 0=normal 1=performance analysis],	Indicates the tachograph performance; including electronic or mechanical analysis, instrument analysis, speed sensor analysis, mass storage analysis, and printer analysis. 00 - Normal performance 01 - Performance analysis
[Tachometer direction 0=forward 1 =reverse],	Indicates the direction of the vehicle. 00 - Forward 01 – Reverse
[event]	
<CR><LF>	

The CANBUS/OBD reports stored in the memory can be request from the OTA commands.

For example REQUEST CANBUS/OBD REPORT:

&&[ID],C99,[CAN report type],[start],[end] [CR][LF]

[CAN report type]: 0 -> ALL CANBUS/OBD report; 1 -> report T1; 2 -> report T2; 3 -> report T3; 4 -> report T4; 5 -> report T5; 6 -> Stop sending CANBUS/OBD report

[start],[end]: request CAN report in-between this period

[start] -> start time

[end] -> end time

The string contains the date and time from CANBUS/OBD

The format is: Year month day hour minute second

For example:

The string 040315131415 means: The date is "Year 2004, March, Day 15 ", and the time is "13:14:15"

If [start] and [end] are empty > then it means request CURRENT CANBUS/OBD report.

1. CANBUS/OBD trigger reports

The UNIT will send the trigger report about CANBUS/OBD to the server when the preset CANBUS/OBD event is occurred.

The packet of CANBUS/OBD trigger report will use the standard report format as below:

**\$EXT,[ID],[GPS Valid],[Date & Time],[Loc],[Speed],[Dir],[Temp],[Fuel],[number of satellite],[GSM signal],
[mileage],[ID Card number],[Status],[Event],CFG:[Message][CR][LF]**

Portman GPRS tracking Event code

Event code	Application
1 ~ 299	The tracking units actively send reports to server.
1001 ~ 2999	
300 ~ 399	Feedback of X command from server.
401 ~ 499	Feedback of Y command from server.
501 ~ 599	Feedback of Z command from server.
701 ~ 799	Feedback of ZE command from server.
801 ~ 899	The tracking units actively send reports to server for CANBUS & OBD functions.
3001 ~ 3099	Feedback of C command from server.
3201 ~ 3299	Feedback of RE command from server.

Event code	Report Description
41	Add fuel report
42	Steal fuel report
58	Water invade fuel sensor
60	Fuel sensor bad
84	Turn-over repoprt
85	Drift report
101	Over the defined maximum speed limitation
102	Low fuel report
103	Course change (direction change) bigger than defined angles.
104	Exit defined Circular Geo-fence area0
105	Enter defined Circular Geo-fence area0

106	Exit defined Rectangular Geo-fence area0
107	Enter defined Rectangular Geo-fence area0
108	Fix time report
109	Fix distance report
111	Low battery report
112	power cut
114	Exit defined Circular Geo-fence area1
115	Enter defined Circular Geo-fence area1
116	Exit defined Rectangular Geo-fence area1
117	Enter defined Rectangular Geo-fence area1
118	Unit Wake up report. (When the unit is back to working status.)
124	Exit defined Circular Geo-fence area2
125	Enter defined Circular Geo-fence area2
126	Exit defined Rectangular Geo-fence area2
127	Enter defined Rectangular Geo-fence area2
130	Power of wireless door sensor1 is low
131	Power of wireless door sensor2 is low
132	Power of wireless door sensor3 is low
133	Power of wireless door sensor4 is low
134	Exit defined Circular Geo-fence area3
135	Enter defined Circular Geo-fence area3
136	Exit defined Rectangular Geo-fence area3
137	Enter defined Rectangular Geo-fence area3
138	tow report
139	Tamper switch of temperature sensor is open
140	Tamper switch of temperature sensor is close

141	Power of temperature sensor is low
142	Stopped report
143	Crash trigger report
144	Exit defined Circular Geo-fence area4 (Self Geo-fence)
145	Enter defined Circular Geo-fence area4 (Self Geo-fence)
146	Exit defined Rectangular Geo-fence area4
198	Coat lose report
150	Press Valet switch to use 'Help' button trigger
156	Door Close (Wireless Door Sensor 1)
157	Door Open (Wireless Door Sensor 1)
158	Door Close (Wireless Door Sensor 2)
159	Door Open (Wireless Door Sensor 2)
160	Door Close (Wireless Door Sensor 3)
161	Door Open (Wireless Door Sensor 3)
162	Door Close (Wireless Door Sensor 4)
163	Door Open (Wireless Door Sensor 4)
166	Tamper Switch Close (Wireless Door Sensor 1)
167	Tamper Switch Open (Wireless Door Sensor 1)
168	Tamper Switch Close (Wireless Door Sensor 2)
169	Tamper Switch Open (Wireless Door Sensor 2)
170	Tamper Switch Close (Wireless Door Sensor 3)
171	Tamper Switch Open (Wireless Door Sensor 3)
172	Tamper Switch Close (Wireless Door Sensor 4)
173	Tamper Switch Open (Wireless Door Sensor 4)
199	Air download is failed
250	Save flash error
237	wireless SOS button is power low

251	Input 1 (in-vehicle control INPUT1 / Door) Door open
252	Input 1 (in-vehicle control INPUT1/ Door) Door close
253	Input 2 (in-vehicle control INPUT2/ACC) ACC on
279	Unit sends message from MDT
258	(TL sensor/Shock sensor/UL sensor) be triggered
268	ID Card information report
269	Invalid ID Card information report
276	Update ID card report
280	Enter defined Point Geo-fence area 0,
281	Enter defined Point Geo-fence area 1
282	Enter defined Point Geo-fence area 2,
283	Enter defined Point Geo-fence area 3,
284	Enter defined Point Geo-fence area 4,
285	Enter defined Point Geo-fence area 5,
286	Enter defined Point Geo-fence area 6,
287	Enter defined Point Geo-fence area 7,
288	Enter defined Point Geo-fence area 8,
289	Enter defined Point Geo-fence area 9,
290	Enter defined Point Geo-fence area 10,
291	Enter defined Point Geo-fence area 11,
292	Enter defined Point Geo-fence area 12,
293	Enter defined Point Geo-fence area 13,
294	Enter defined Point Geo-fence area 14,
295	Enter defined Point Geo-fence area 15,
296	Enter defined Point Geo-fence area 16,
297	Enter defined Point Geo-fence area 17,
298	Enter defined Point Geo-fence area 18,

299	Enter defined Point Geo-fence area 19
301	Door lock (in-vehicle output) command Success
302	Door unlock (in-vehicle output) command Success
303	ARM (in-vehicle output) e command Success
304	Panic mode (in-vehicle output) command Success
305	Anti-Car jacking (in-vehicle output) command Success
306	Emergency Release (in-vehicle output) command Success
307	Trunk release (in-vehicle output) command Success
369	Adjust OD100
401	Request current location
402	Wake up the UNIT to go online / connect to GPRS and server.
403	Let the UNIT go off line in GSM mode.
404	Let the system temporarily report in [connection time] duration and using the [report rate] to send the report
405	Clear all reports in the memory
406	Read the UNIT's firmware version.
407	Check the UNIT Geo-fence status.
408	Read external battery level
419	Clear all tire pressure Sensor
412	Set the preset distance
432	Set Self-geofence
435	reset the GPS module
436	reset the unit
483	Control buzzer chirp
499	Active tracking command
501	Setup the dial-up interval, reconnect times and GPRS dial-up mode
502	Setup the Circular Geo-fence or Immediate Geo-fence parameters

503	Setup the Rectangular Geo-fence parameters
504	Ask for history report by certain index range
507	Set SMS number and clear date
508	Setup sleep mode parameters
509	Setup Point Geo-fence
510	Setup APN, user name, and password
511	Setup dial-up procedure parameters
512	Keep alive parameter
513	Setup device id, device password
516	Setup inside battery work
520	Setup sensor parameters
521	Set up report after successfully keeps alive ACK
524	Setup Seven Input
525	Setup FN700 send Add fuel or Steal fuel report
530	Setup fixed distance service report
529	Setup low fuel report
532	Setup Intelligent report
533	Setup Intelligent history report
535	Setup use OD100 or not
536	Setup the fixed distance report
537	Setup Course change report
538	Setup Temperature report
539	Remote setup server IP and port
543	Setup intelligent report for SBD
578	Set Report for code learn mode to Enable or Disable
547	Setup the id-card report

548	Setup the id-card immobilization function
549	Setup using id-card or not
573	Set steal fuel report
574	Set add fuel report
575	Set ID card parameters
579	Set Report for INPUT1 (In-vehicle DOOR) changes to Enable or Disable
597	ACK for the unit receiver the command message for MDT
730	Address route navigation
713	Set fuel sensor ID number
732	Setup fuel sensor type
733	Setup wired or wireless temperature sensor
734	Setup wireless door status
745	Setup trailer report
860	Fixed time report (for CANBUS/OBD T1 & T4 reports)
861	Trip report while ACC off (for CANBUS/OBD T2 & T3 reports)
862	Tachometer report while status changed (for CAN50 T5 report)
870	Mover running report
871	Normal Fuel decrease report > when fuel is changed in __% and odometer changed is less than __ km (default: 5%, 1 km)
872	Fuel consumption report > when instantaneous fuel consumption is over __ L/100km (default: 7L/100km)
873	Engine idle time report > when engine is idle for over ____ seconds (default: 3mins)

874	Engine over heat report> when engine temperature is over ___ C (default:150C)
875	Engine running time report> ___ when engine running time is reach ___ hr/days (default: 90 days)
876	Over pressing fuel pedal report > when press fuel pedal is over ___ % for ___ seconds (Default: 95%, 5 seconds.)
877	Emergency pressing brake pedal report > when press brake pedal is over ___ % for ___ seconds, when the vehicle speed is over ___ km/h, Default: 95%, 5 seconds. 50km/h
878	Driver work overtime notice report> ___ Send report when driver is continues working more than ___ hours, default: 5 hrs
879	Engine malfunction indicator / diagnostic report: (not include in FMS, only in J1939) > Send report when the following lamp status is detected: Malfunction indicator lamp on
881	Duty On report (For CAN50 Only)
882	Duty Off report (For CAN50 Only)
883	DTC Remained
988	Fuel sensor Removed
998	external car alarm type
999	External alarm report
3004	Setup recording CANBUS/OBD report
3005	Setup sending CANBUS/OBD report
3007	Setup RPM over running report
3008	Setup Abnormal Fuel decrease report
3009	Setup Fuel consumption report
3010	Setup Engine idle time report
3011	Setup Engine over heat report
3013	Setup Over pressing fuel pedal report
3014	Setup Emergency pressing brake pedal report
3015	Setup Driver overtime notice report
3016	Setup Malfunction indicator lamp on
3017	Setup duty on/off report